

MIC501T- Microbiology (Theory) **Old Code # BT 102**

MID TEM PPT (001 – 52)

FINAL TERM PPT (53 – 104)

Merge by BAQIR HUSSAIN

BS ZOOLOGY 8th Semester

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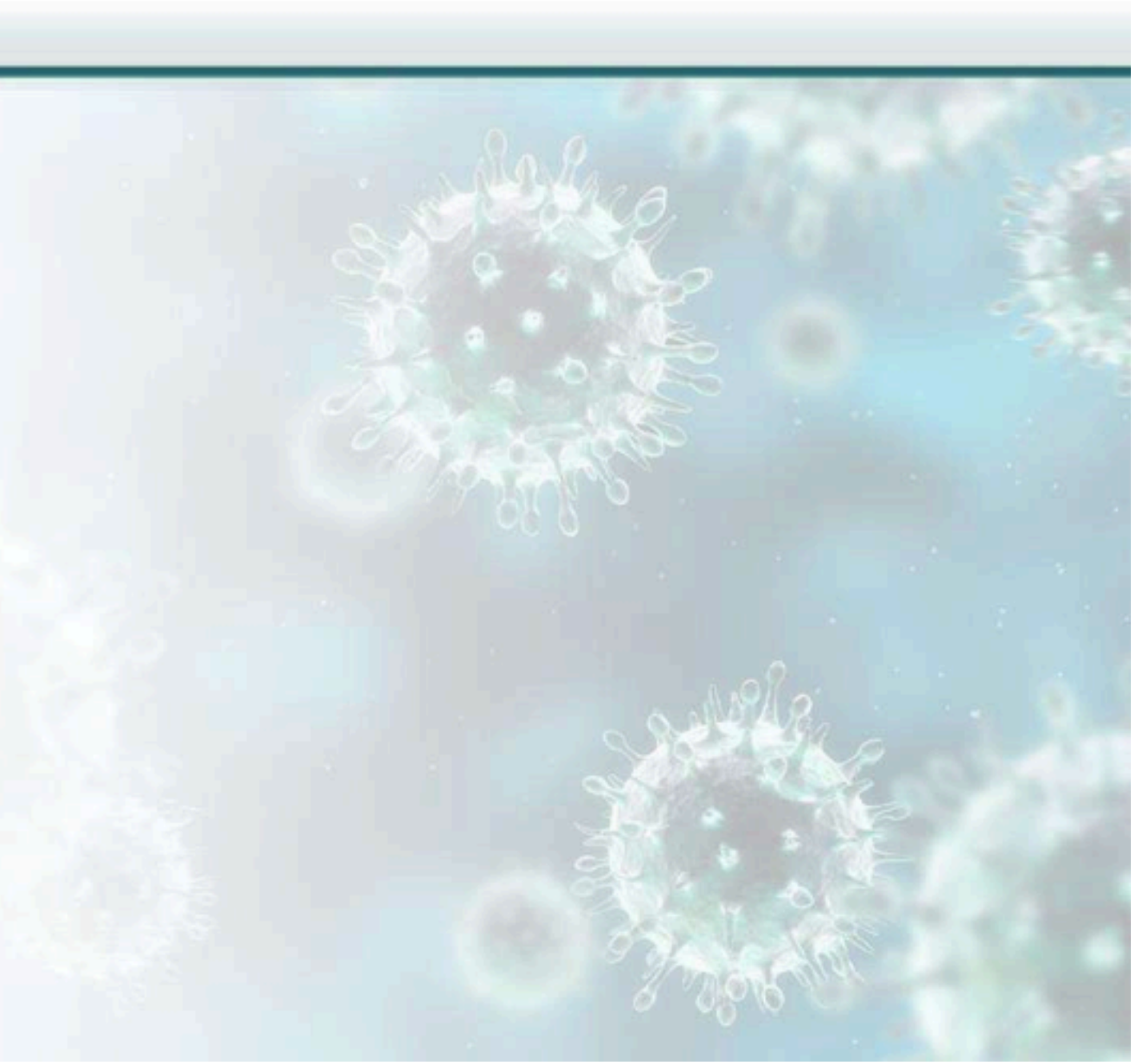


Microbes and

Microbiology

Module # 001

- **Learning Objectives**
- What are microbes?
- What is meant by microbiology?

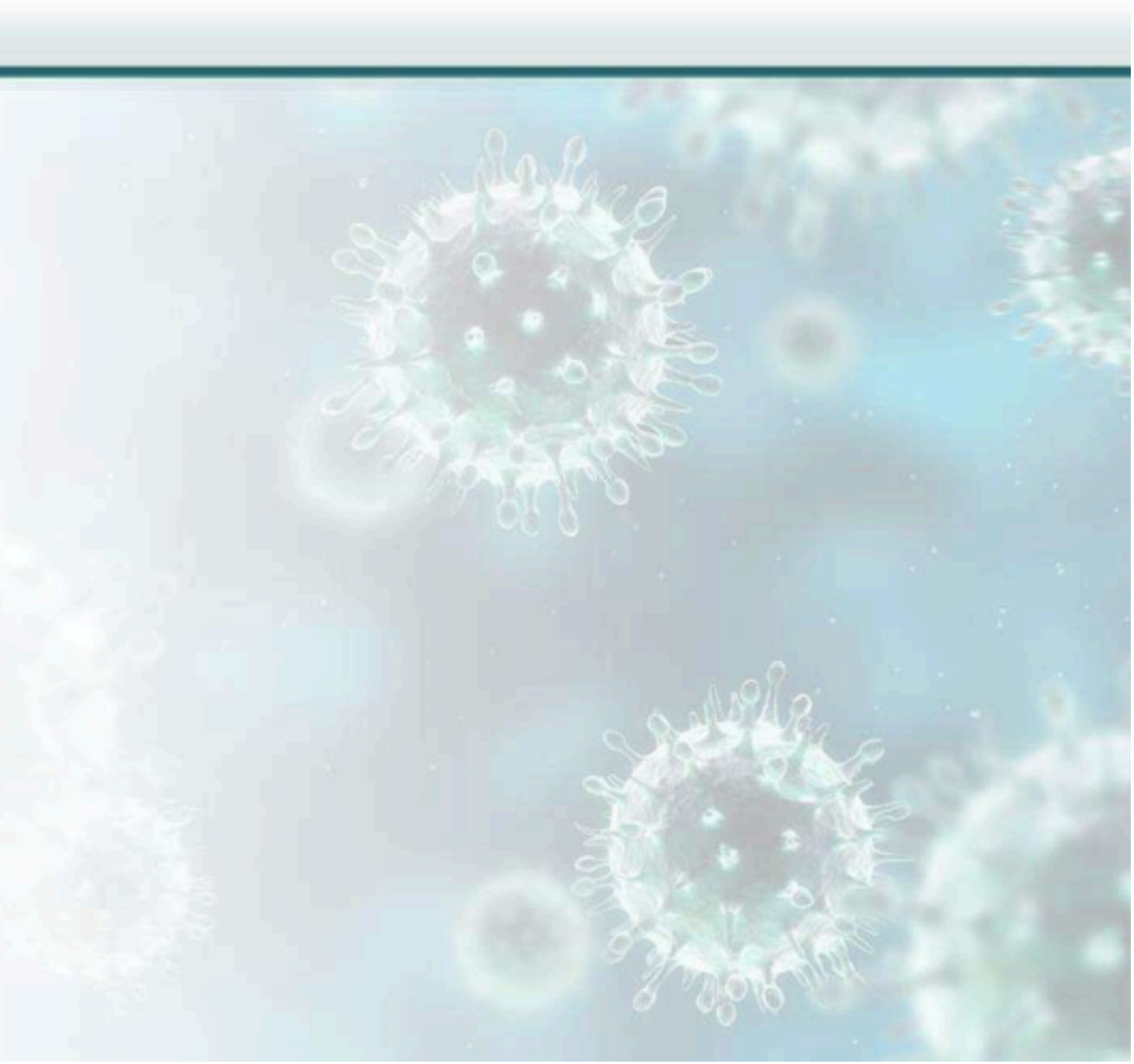


What are Microbes?

- Also called germs or

microorganisms

- Too small to be seen with the unaided eye
- A microscope is needed to see them
- Studying microbes is microbiology

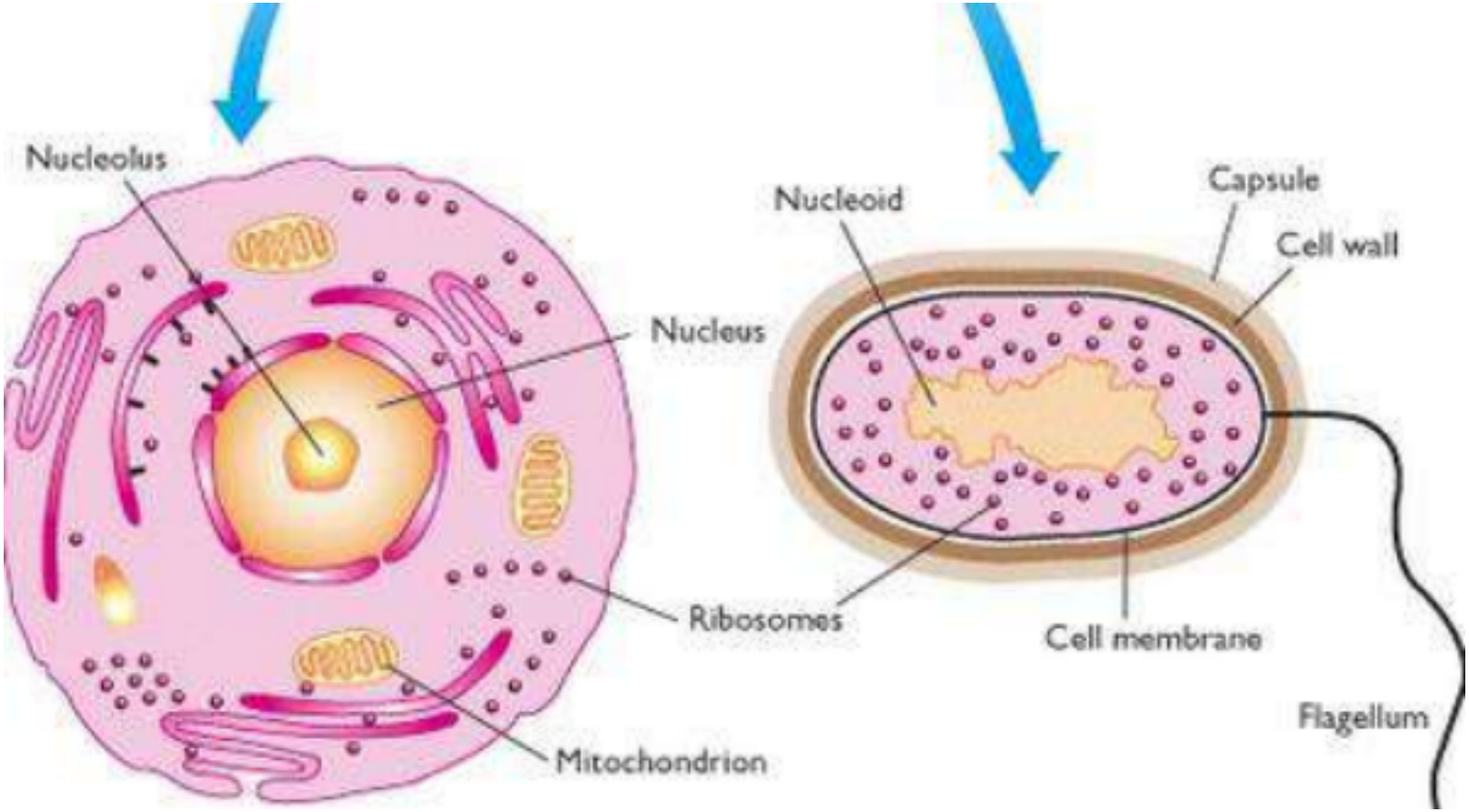


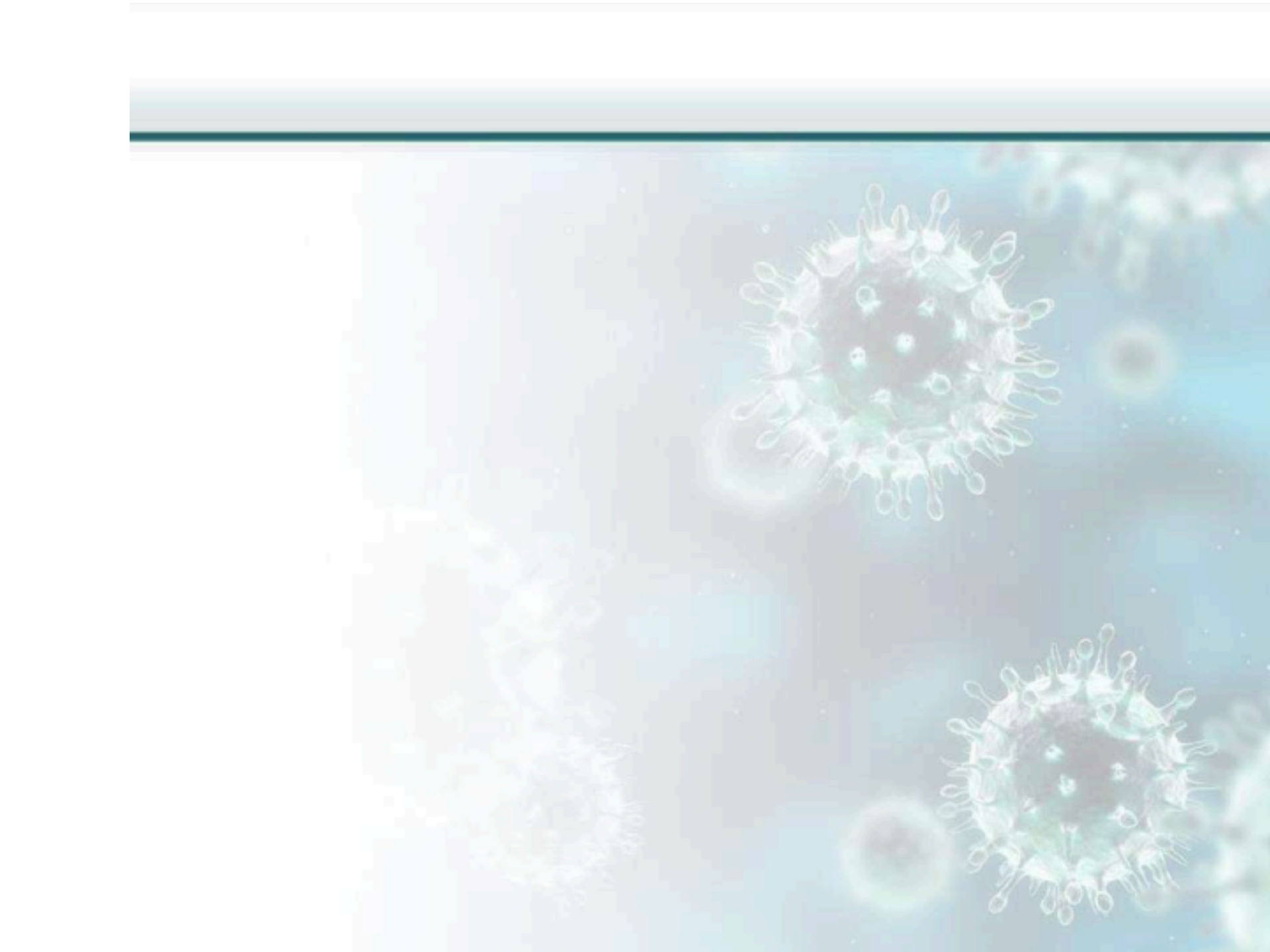
What are Microbes?

Microbes are simple.

- Diverse
- Classification has been a challenge.

- Procaryotes (no nuclear membrane) and eucaryotes (distinct nucleus)





Eucaryotic Vs Procaryotic Cell

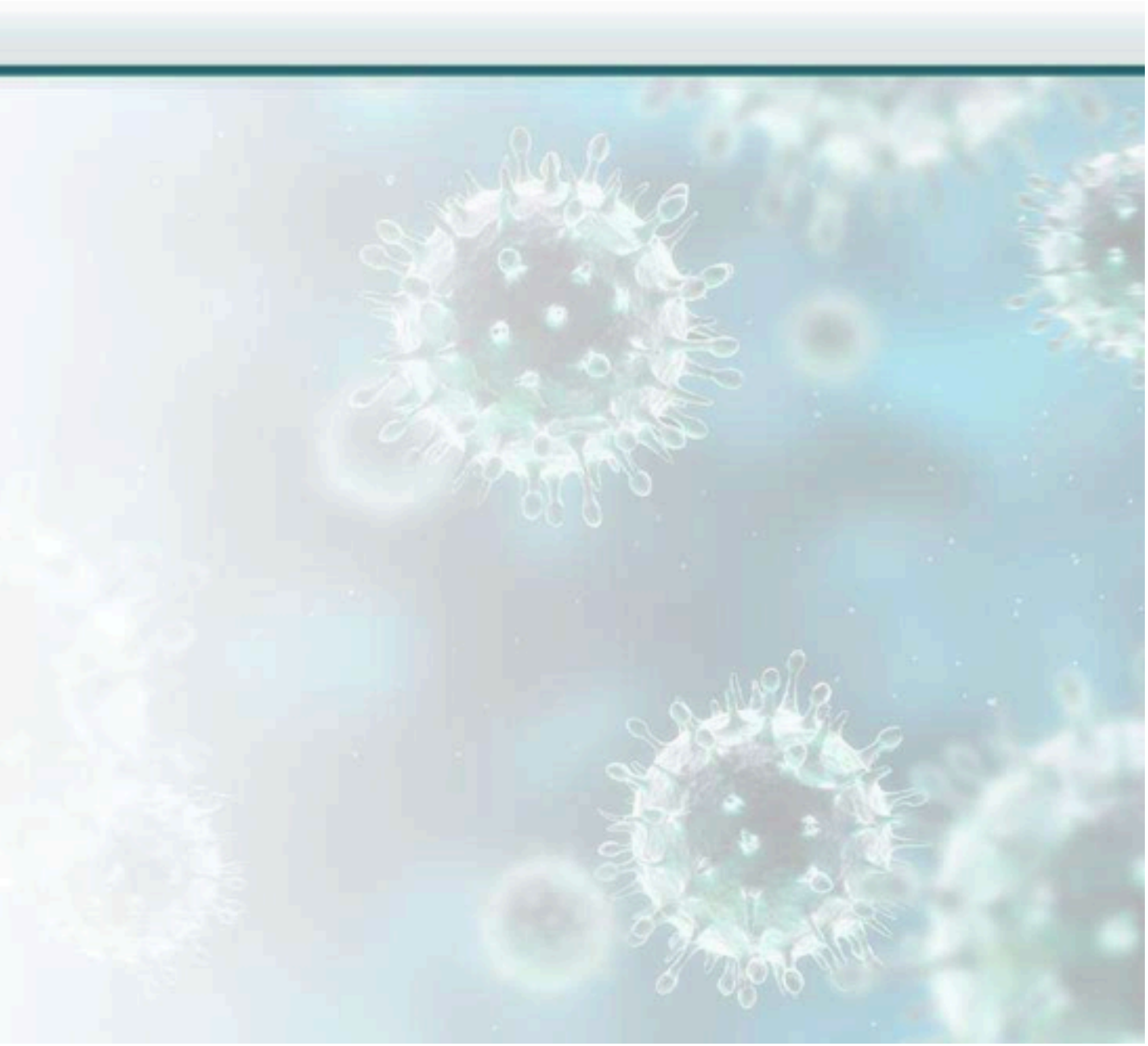


Why do we Classify

Organisms?

Module # 002

- Studying them becomes convenient
- Easy Identification
- Easy to establish relationship between organisms
- Evolutionary relationships can be established



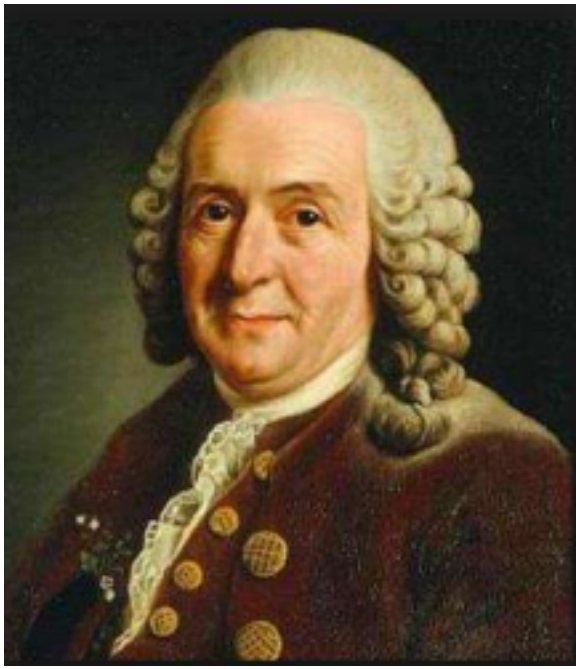
Bases of

Classification

- **Similarities among them**
- Morphology used to be the main feature prior to the advent of molecular biology
- Biochemical similarities
- Genetic similarities
- Evolutionary relationships



Five Kingdom Classification of Organisms



Carl Linnaeus, 1735

- Monera: Bacteria and Archea
- Protista
- Fungi
- Plantae
- Animalia

- Microbes placed in the first 3 kingdoms

Carl Linnaeus, 1735

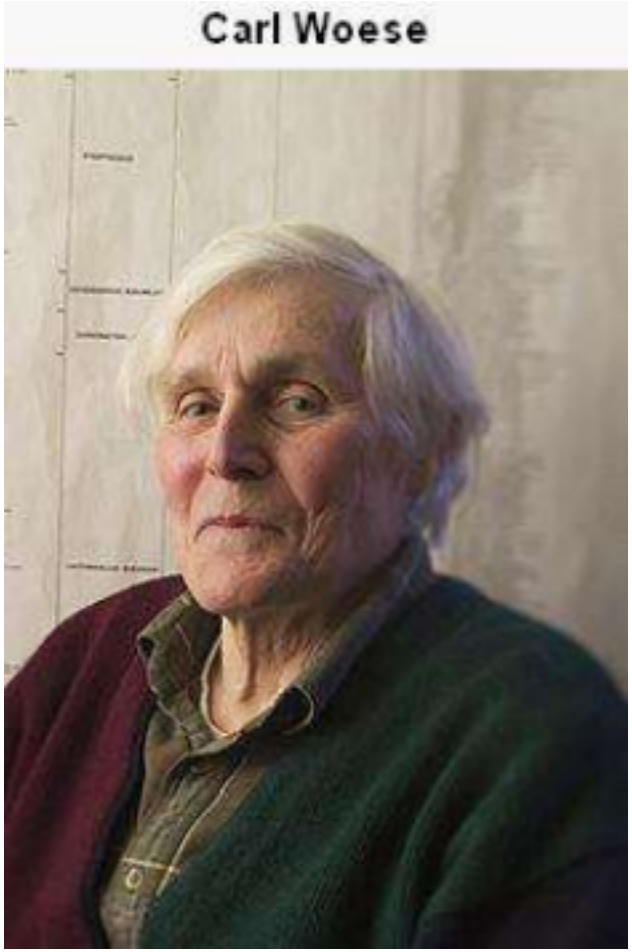


Reasons for Changes in the Classification

- Electron microscope
- Biochemical and physiological characterstics
- Nucleic acid and protein sequencing
  - Ribosomal RNA (rRNA)
- Bacteria and Archea two different groups



Three Domain Classfication (1977)



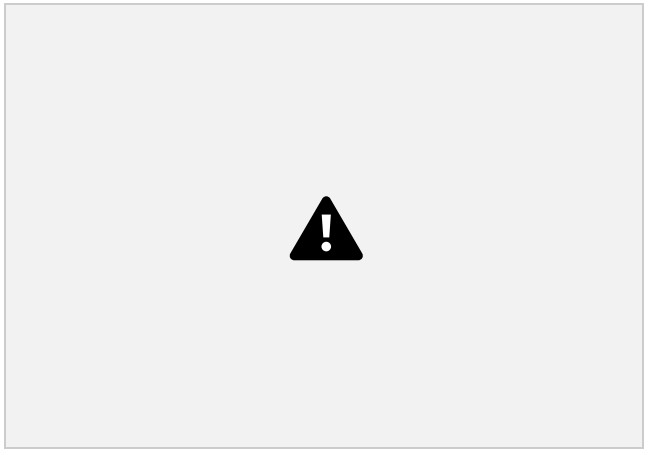
- Bacteria: True or eubacteria
- Archea:
- Eucarya
  - Protists (slime molds, protozoa, algae)
  - Fungi
  - Plants
  - Animals



Domains &

Kingdoms

- Domain: Bacteria, Kingdom Bacteria
- Domain: Archea, Kingdom Archea



- Domain: Eucarya
  Kingdom Protista
  Kingdom Fungi
  Kingdom Plantae
  Kingdom Animalia



Bacteria

- Procaryotes
- Usually single-celled organisms
- Cell wall made up of peptidoglycan



- Procaryotes
- Cell wall has no peptidoglycan, but has special lipids.
- Vary in rRNA sequencing
- Live in extreme conditions – Thermophiles – Halophiles

Archea



Eucarya

Protists

- Fungi
- Animals
- plants

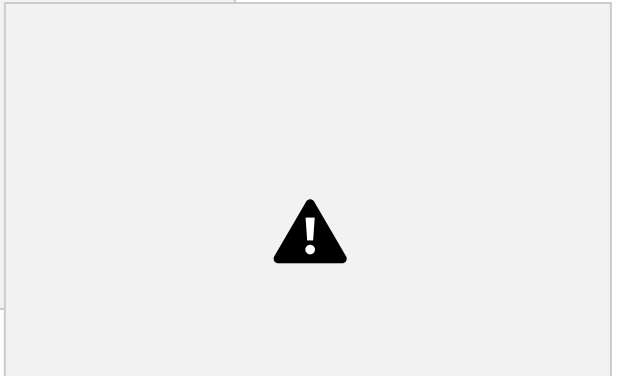


- **Algae:**



Cellulose cell wall, Photosynthetic Produce (O_2)

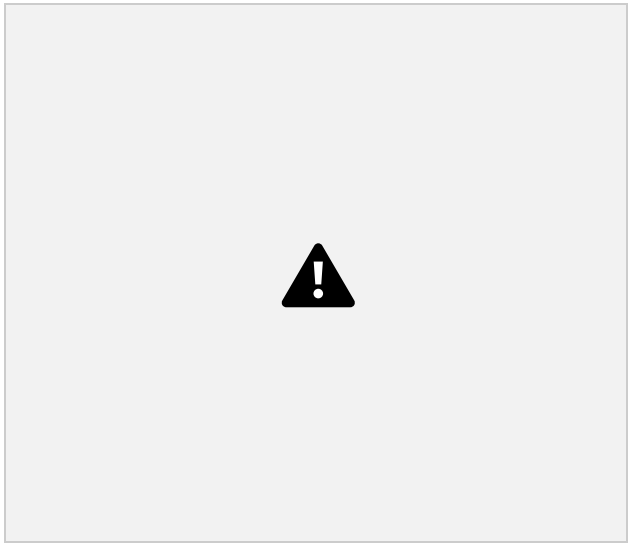
- **Protozoa:** motile, unicellular organims



- **Slime mold:**

Protists

Protozoa like at one stage fungus like in another and





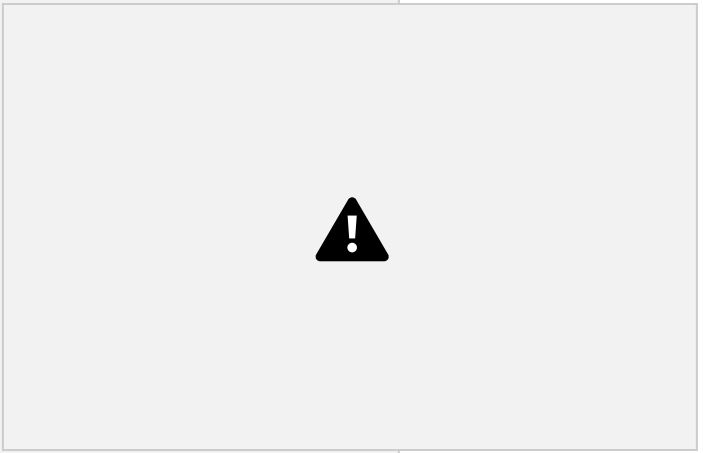
Fungi

- Unicellular form is called yeast.
- Multicellular is known as mold or mushroom

• Viruses: Acellular organisms

– Proteins and nucleic acids





• Viroids: composed only of circular ssRNA

Other Microbes

(potato
spindle tuber viroid)

- Virusoids: Contain circular ssRNA, need helper viruses

for replication and encapsidation; also called satellite viruses

- Prions: Infectious proteins



Naming Organisms: Binomial Nomenclature

- **Linnaeus** (in 1735) established the system of scientific nomenclature
- Each organism has two names: the **genus** and **specific epithet** (descriptive word)
  - *Escherichia coli*



- Are italicized or underlined. The genus is capitalized, and the specific epithet is lowercase.
- Are "Latinized" and used

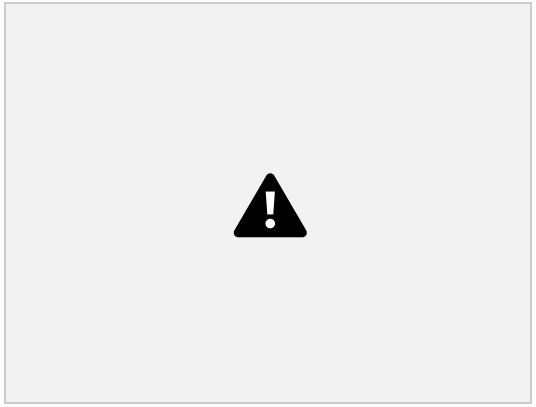
worldwide.

- May be descriptive or honor a scientist.



- Honors the discoverer, Theodor Escherich
- Describes the bacterium's habitat—the large intestine, or colon

Escherichia coli





Staphylococcus aureus

Describes the clustered (*staphylo-*) spherical (*cocci*) cells

- Describes the gold colored (*aureus*)

After the first use, scientific names may be abbreviated with the first letter of the genus and the specific epithet:

- *Escherichia coli* and *Staphylococcus aureus* are found in the human body. *E. coli* is found in the large intestine, and *S. aureus* is on skin.



Bacterium Vs

Bacteria

- E.coli as a species is a bacterium.
- One E.coli cell is a bacterium. • Many E.coli cells are bacteria. •

Medium Vs Media



History of

Microbiology

Module # 003

Learning Objectives

- First microscopic observation of microorganism

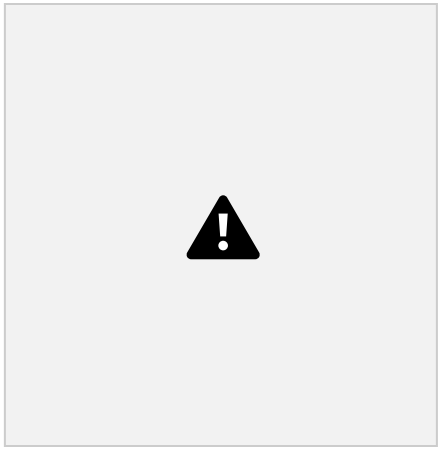


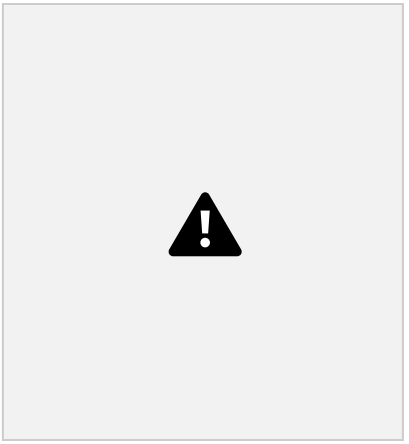
History of

Microbiology

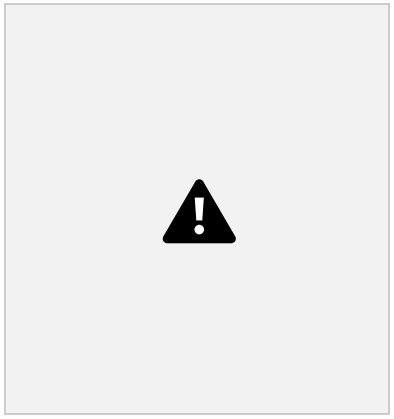
Discovery of Microbes

- Investigators suspected existence of microbes.





Lucretius Fracastoro

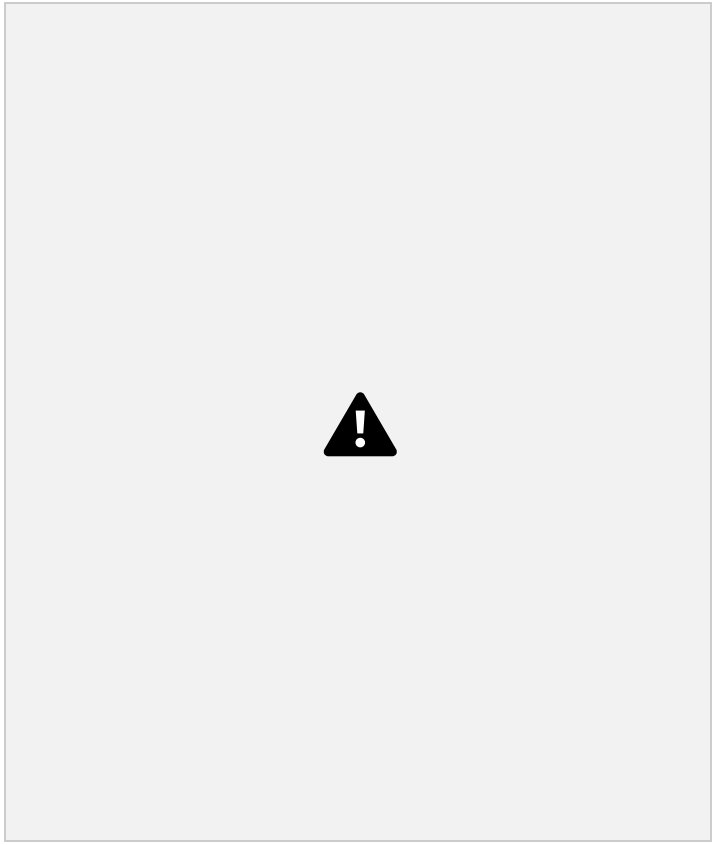


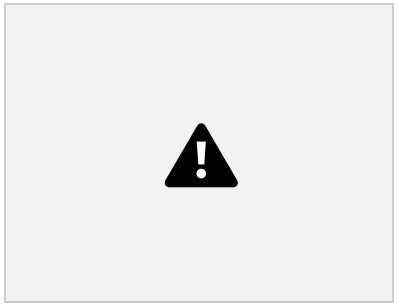


History of Microbiology

- First published drawing of microbes in 1665.
- Man behind the cell theory.

Robert Hooke







History of Microbiology

Antony van Leeuwenhoek

- First to publish

extensive and accurate observations of microbes

- He called them "animalcules".



History of

Microbiology

Module # 004

Learning Objectives

- What was spontaneous generation?
- How was spontaneous generation issue solved?

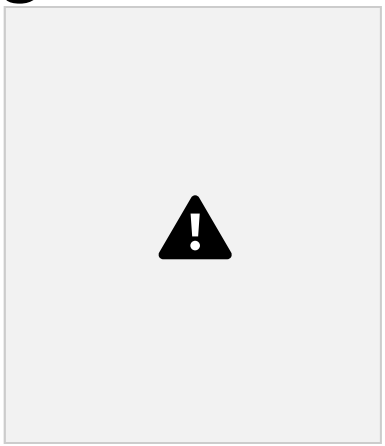


History of

Microbiology

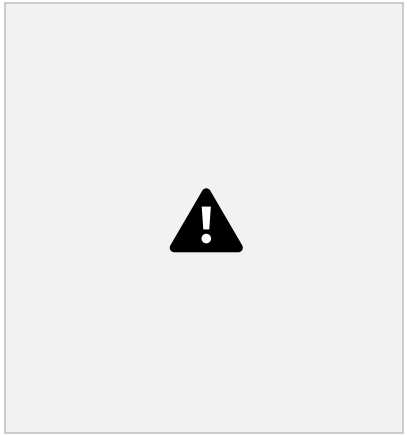
arise
by spontaneous
generation

Spontaneous Generation



- Aristotle: Simpler

invertebrate could



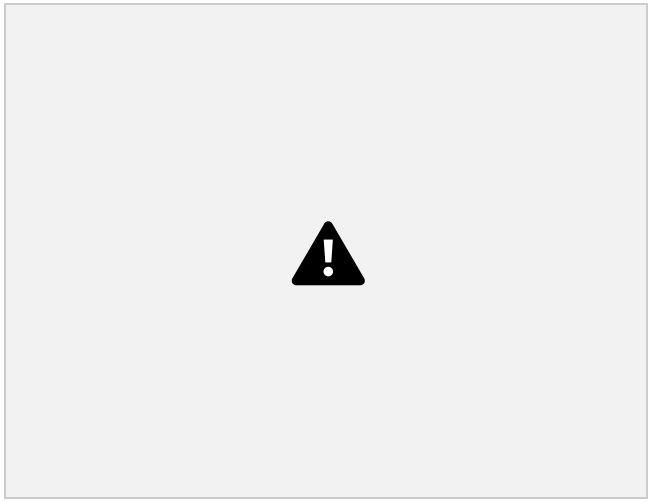


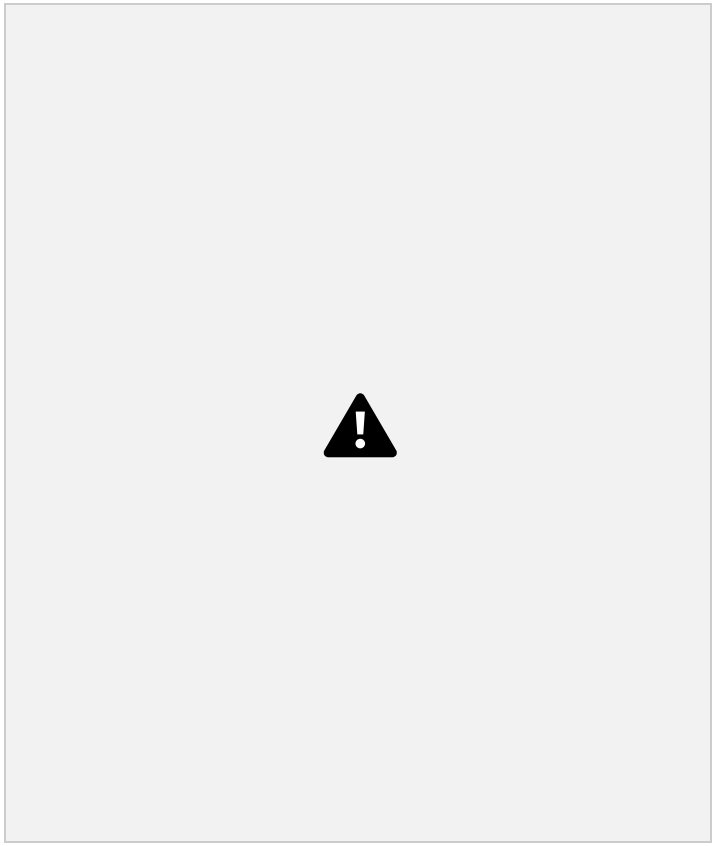


History of Microbiology

Spontaneous Generation

- Francesco Redi challenged







History of Microbiology

Leeuwenhoek

- His observations of microbes renewed the controversy.



History of

Microbiology

John Needham (1748)

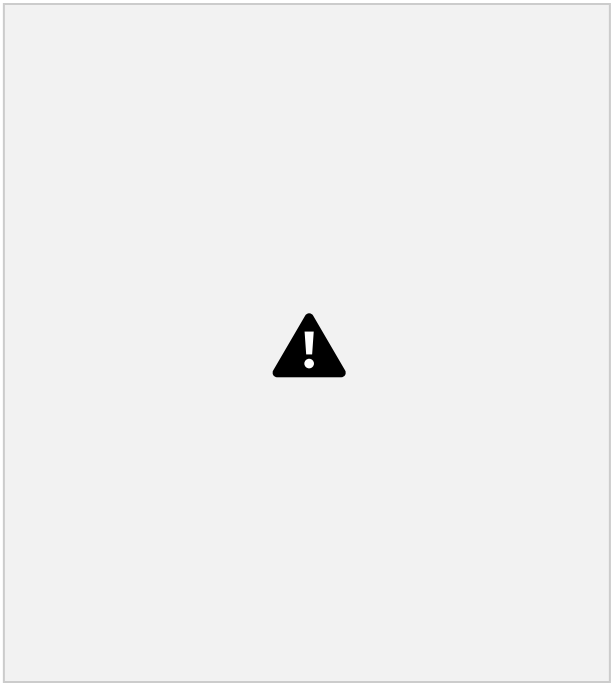
- Boiled extracts of hay or meat can give rise to microorganisms
- He boiled mutton broth and then tightly stoppered the bottles.
- Conclusion: organic matter has vital force.



History of

Microbiology

Spallanzani



- Sealed bottles before heating
- No growth was seen
- Proposed that air carries the germs
- Also commented that air was necessary for vital force.







History of Microbiology

Louis Pasteur (1861)

- Microbes are present in the air. • They can contaminate sterile solutions



History of

Microbiology

- Pasteur's work ended the spontaneous generation controversy.



The Golden Age of

Microbiology

Module # 005

Learning Objectives

- What discoveries were made
in the golden age of microbiology?



The Golden Age of

Microbiology

- In about 60 years,

1857-1914

- A lots of discoveries in

microbiology with Pasteur's work

Microbiology became a real science.

- Spearheaded by Pasteur

and Koch



The Golden Age of

Microbiology

1857-1914

- Disease agents identified
- Role of Immunity established
- Chemical activities of microbes studied
- Improved culturing techniques introduced



The Golden Age of

Microbiology

1857-1914

- Vaccines
- Surgical techniques developed



The Golden Age of

Microbiology

Fermentation and Pasteurization

- Souring of beer and wine
- Pasteur found the yeast that fermented sugar to alcohol
- In the presence of O2, bacteria grow and change alcohol to acetic acid



The Golden Age of

Microbiology Germ

Theory of Disease

- Treatments for many diseases were discovered before Pasteur.
- Causes of the disease were unknown.
- Yeast fermentation by Pasteur was a clue



The Golden Age of

silkworm disease caused by a protozoan.

Germ Theory of Disease

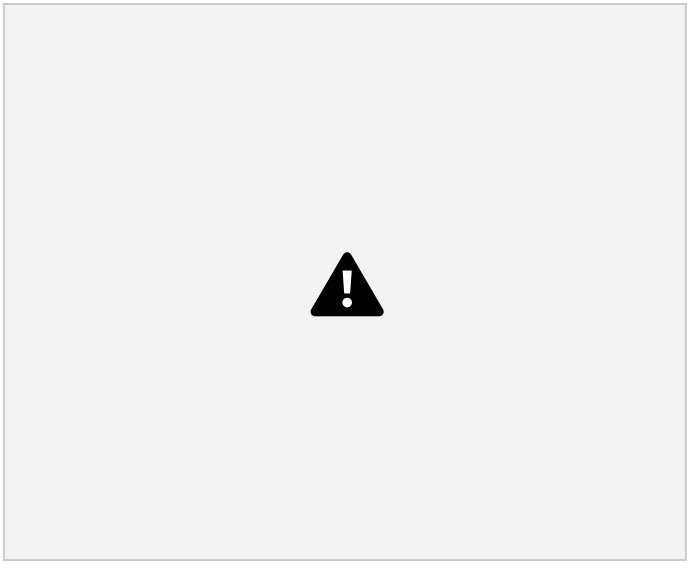
- 1860, Joseph Lister, applied phenol as disinfectants in treating surgical wounds and hand sanitizing.
- In 1865, Pasteur found that



The Golden Age of

Microbiology

Robert Koch



1876: *Bacillus anthracis* established as a cause of anthrax, first proof

- Isolation of bacteria made possible by agar
- and Petri dish





The Golden Age of Microbiology

Koch's Postulates

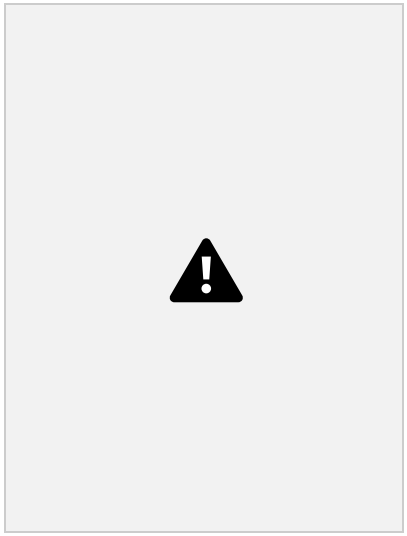


The Golden Age of

Microbiology

Vaccines

- Edward Jenner in 1796 and small pox



- Pasteur in 1888, discovered why vaccination works
- Avirulent microbes as vaccine
- The word “vaccine”



The Golden Age of Microbiology

Ivanowsky

- Viral pathogens
- Tobacco mosaic virus





The Golden Age of Microbiology



Branches of

Microbiology

Module # 006

Learning Objectives

- What are various branches of Microbiology?



Branches of

Microbiology

- We live in the age of bacteria.
- First living organisms on our planet.
- They created the atmosphere that we live in today.
- Biosphere depends upon their activities.



Branches of

Microbiology

– Identification

– Characterization

– Classification

Bacteriology

- Study of bacteria
- Deals with:

– Isolation



Branches of

Microbiology

Mycology

- Study of fungi
- Medical mycology
- Agricultural mycology
- Ecological mycology

– Decomposers of woods, leaves, feces and other organic matters



Branches of

Microbiology

Basic & Applied Microbiology

- Basic: Biology of microbes
- Applied: Practical problems (diseases, wastewater Rx, food spoilage etc)



Branches of

pathogens

Medical Microbiology

- Deals with diseases of humans and animals
- Identifications of disease causing agents
- Control and elimination of microbes
- Tracking down new



Branches of

measures •

Surveillance studies

Epidemiology

- Monitoring diseases in a population
- Detection of outbreaks and developing epidemics
- Implementation of appropriate control



Branches of

Microbiology

Immunology

- Study of the immune system
- How it works
- Allergies
- Vaccines
- Antibodies for the treatment of diseases and cancers



Branches of

Microbiology

Food Microbiology

- Microorganisms that either create or contaminate foods
- Cheese
- Bread
- Yogurt
- Wine



Branches of

Microbiology

Industrial Microbiology

- Use of microbes or their enzymes for large scale production of biomolecules



Branches of

Microbiology

Microbial Genetics

& Mol. Biology

- How genes regulates the development and function of cells and organisms.
- *E. coli, Bacillus subtilis, Saccharomyces cerevisiae,* T4 and lambda phages as study models

Module
007
Learning



Microscopy

Objectives

- What is light microscopy.
- How a light microscope works.
- What are various parts of light microscope.
- What is total magnification.



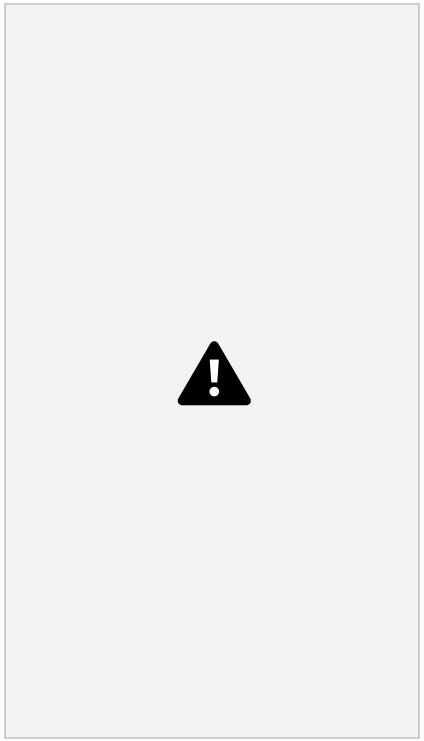
Learning

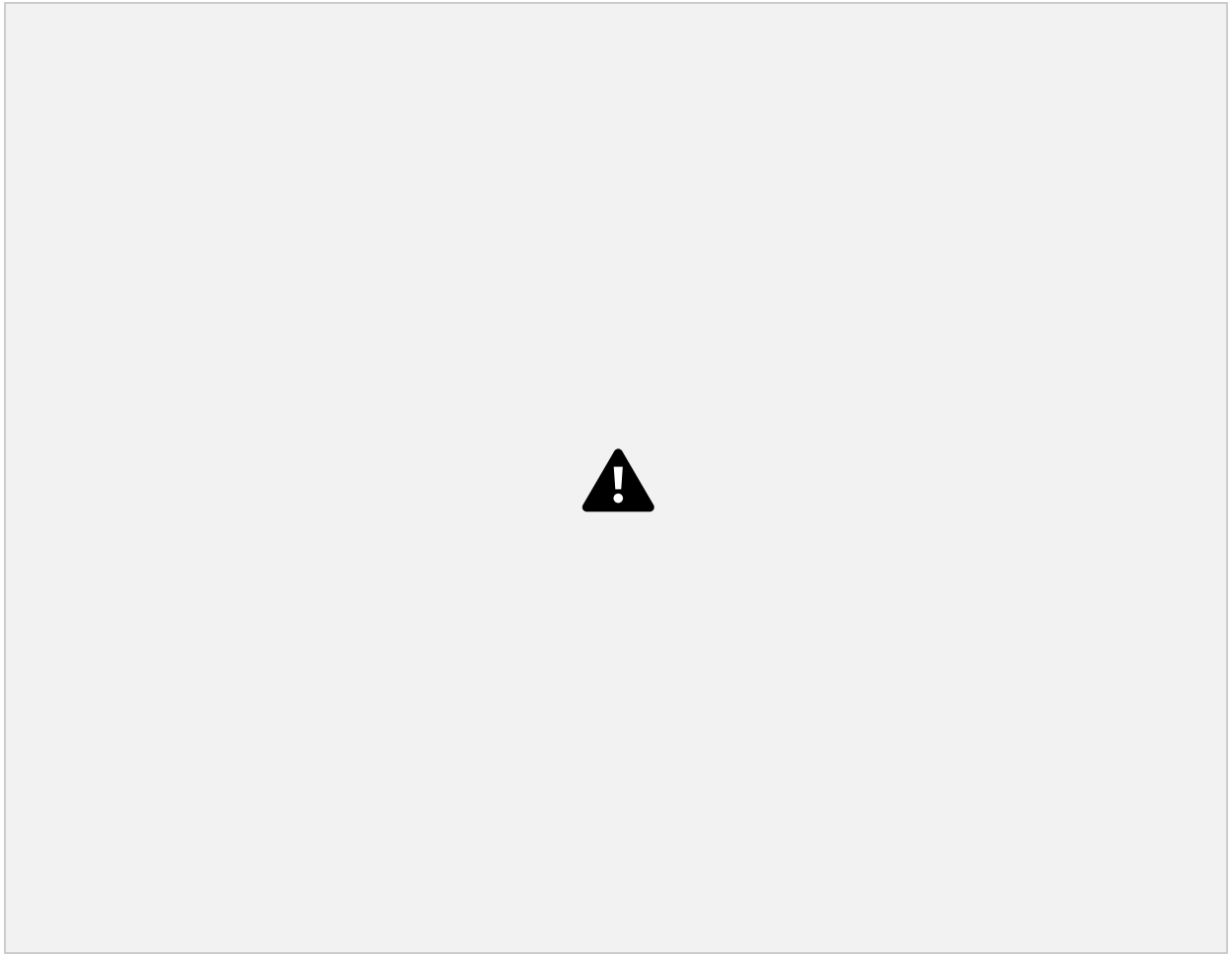
Microscopy

Objectives

refractive index .

- What is resolution and

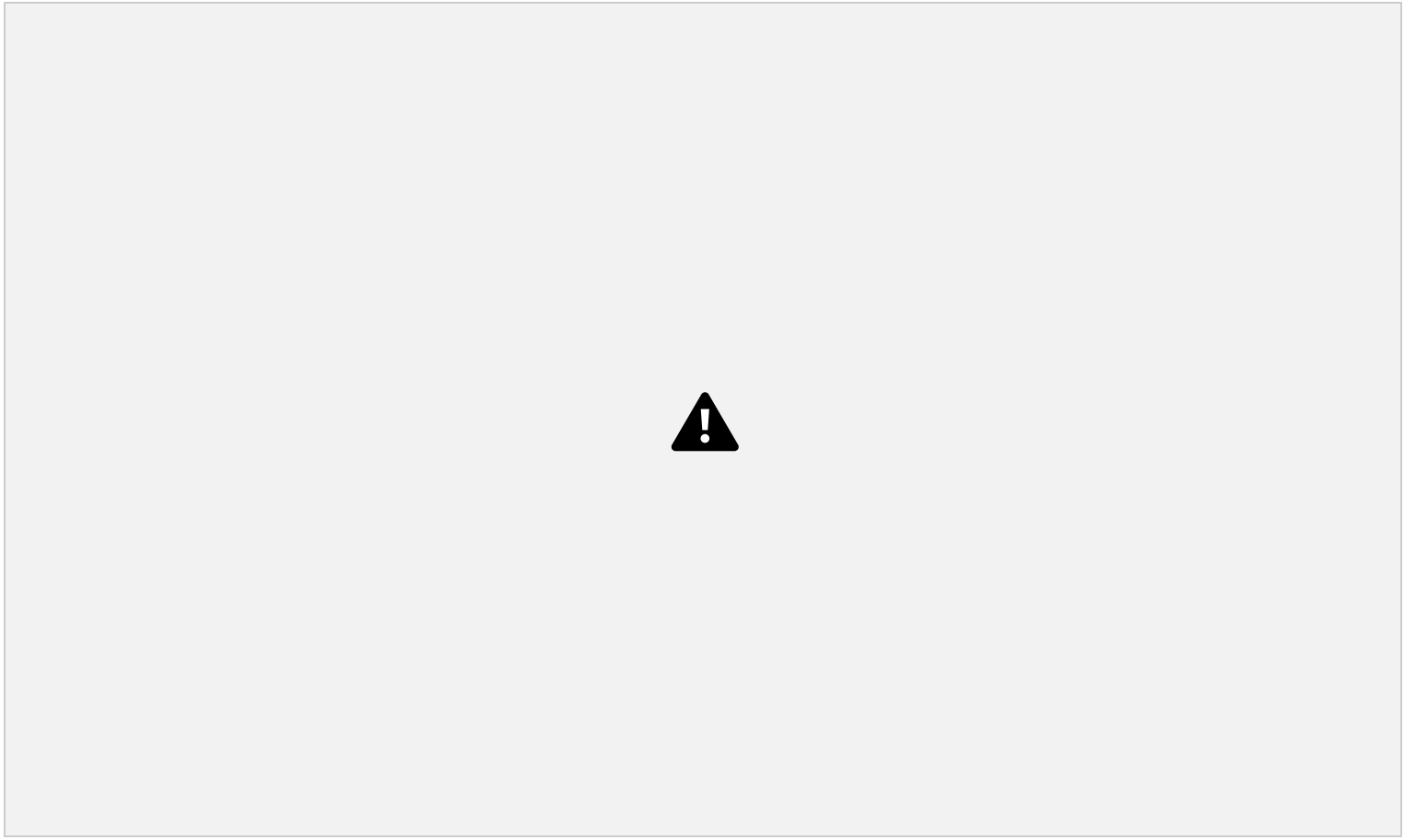






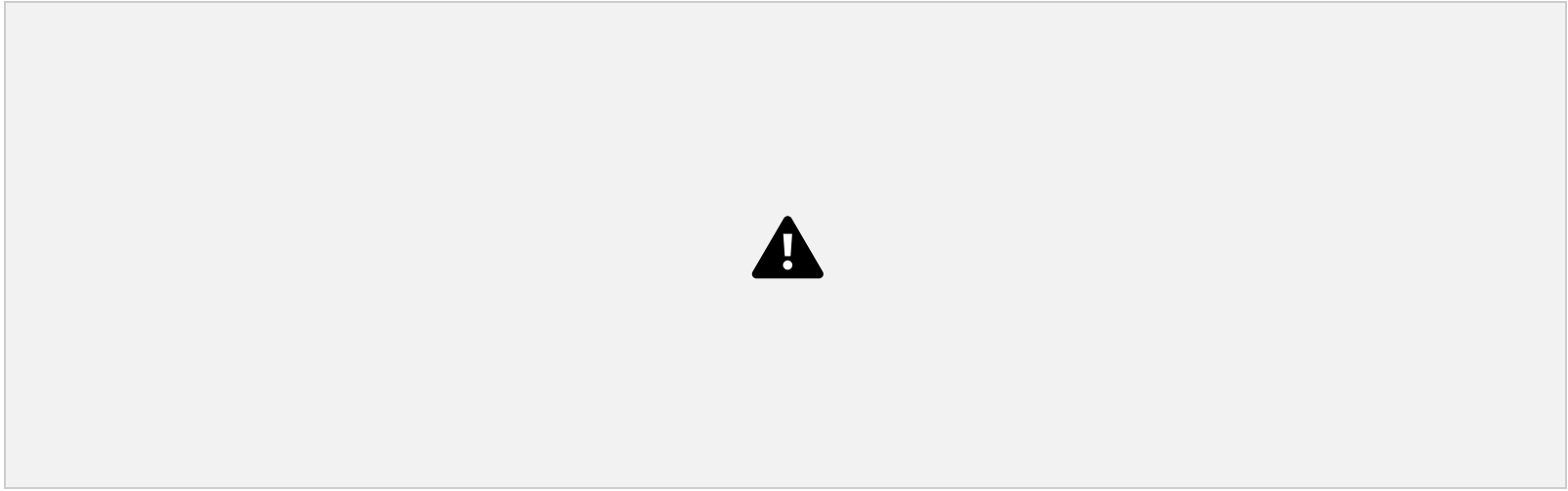
Light Microscope

Uses visible light to observe the specimen



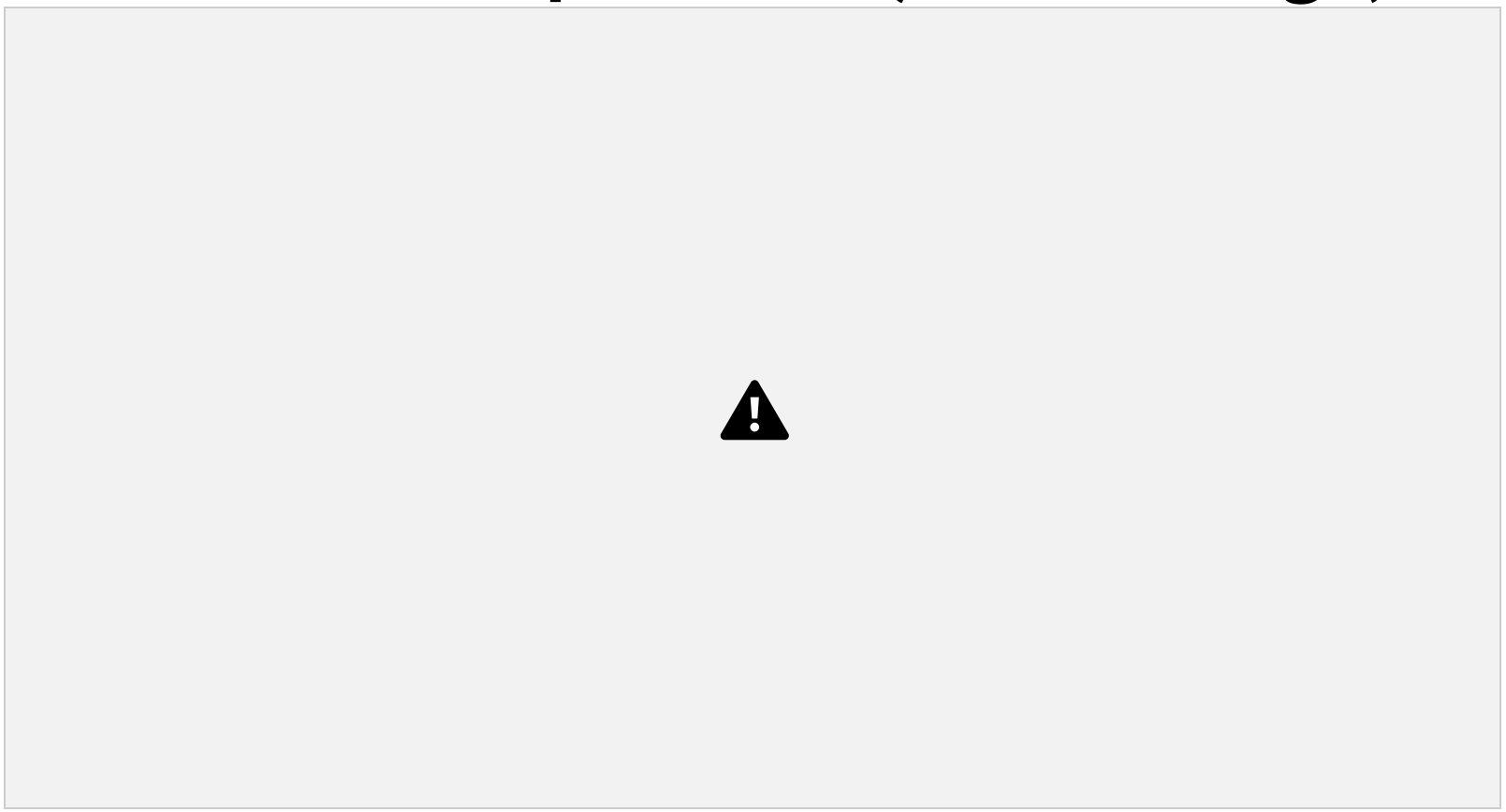


How a microscope works (Real Image)





How a microscope works (Virtual Image)



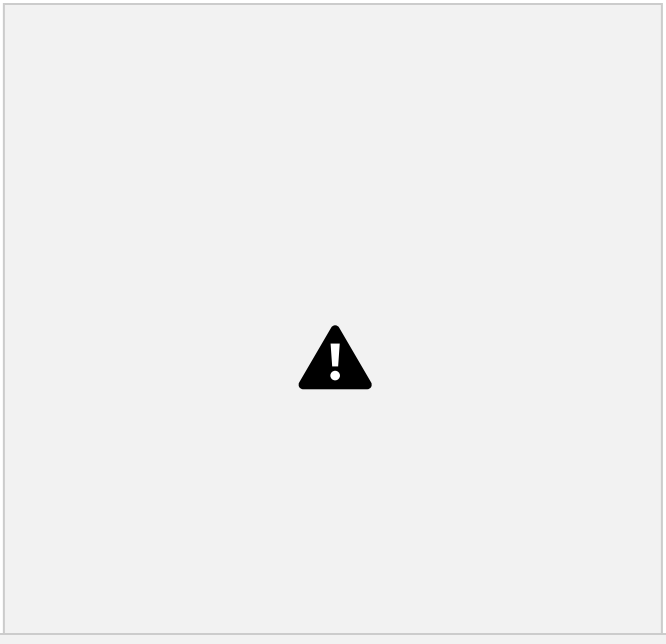


How a microscope works



Parts of Compound

Light Microscope



- Illuminator: a light source
- Condenser: Directs the light through the specimen
- Objective Lenses: Close to the specimen
- Ocular Lens (Eyepiece): Close to the eye

Magnification of objective lens x magnification of ocular lens

- **Objective lenses:**
  - **Low: 10 X**
  - **High: 40 X**
  - **Oil immersion: 100 X**
- **Ocular lens: 10 X**
- **Low magnification power: 10 x 10 = 100**
- **Oil immersion power: 100 x 10 = 1000**

Image

- **Refractive Index:**





Ability of

Refractive Index and

a medium
to bend the light

- Light rays move in a straight line through a single medium

- To increase contrast, we stain the specimen





• Ability of the lenses to distinguish between two closely lying objects as

What is resolution?

separate

- Light microscope resolving power is 0.2 µm.
- n = refractive index, 1.5 for immersion oil



- **Ordinary light microscope**
- **Stained images are used**
- **Can resolve 200 nm objects**

Summary



Module # 008 Learning Objectives

- Darkfield microscopy
- Phase contrast microscopy
- Fluorescent microscopy
-

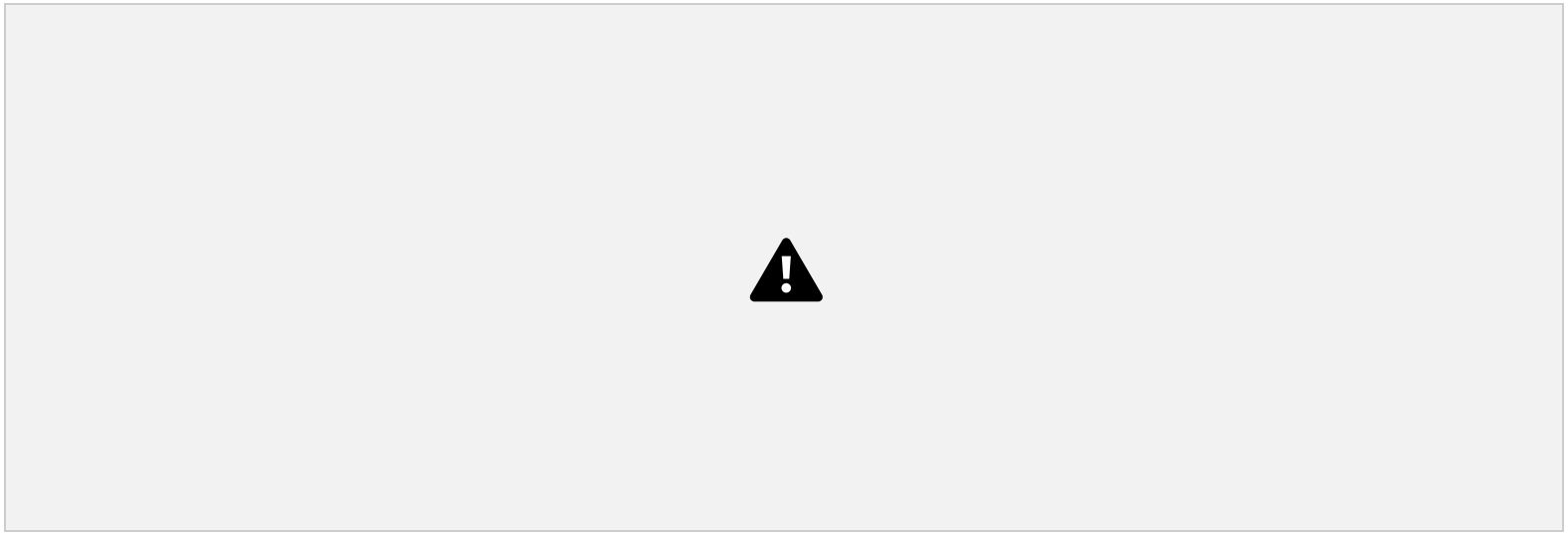
Types of Microscopes

Transmission electron microscopy • Scanning electron microscopy

Darkfield Microscopy

- Uses a darkfield condenser having an opaque disk

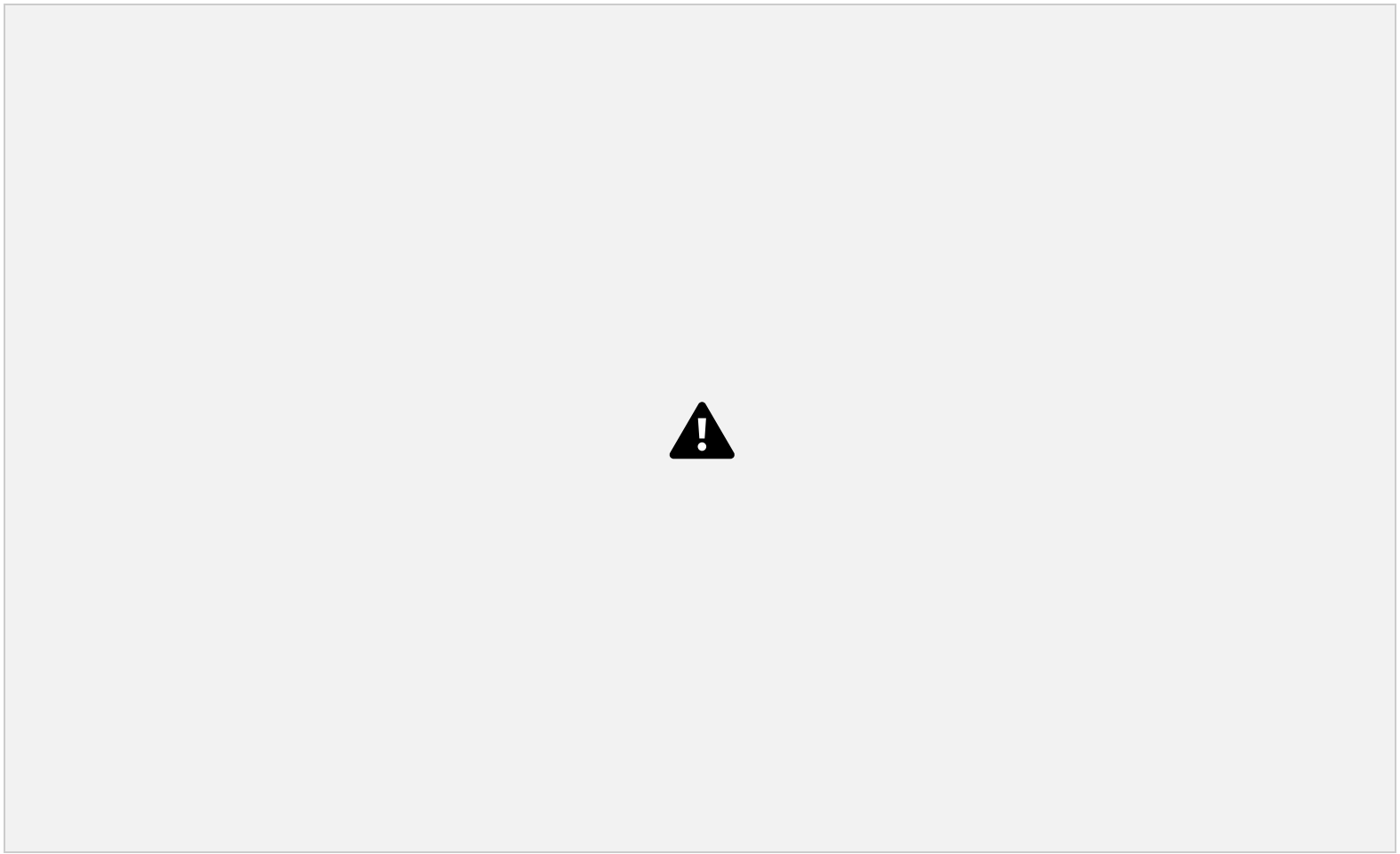
- **Used for unstained specimens suspended in liquids**





Phase Contrast Microscopy

- Light rays can be in phase or
- out of phase





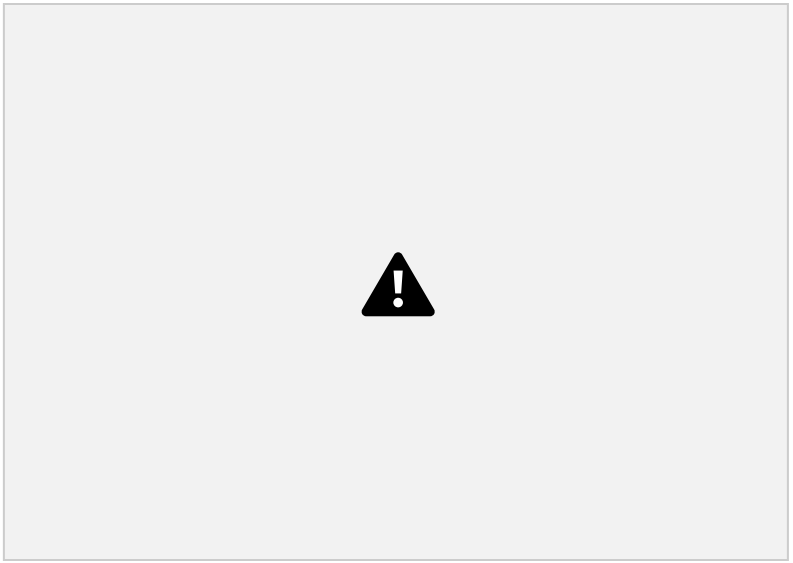
Phase Contrast Microscopy



- Uses

Fluorescent Microscopy

fluorochoromes



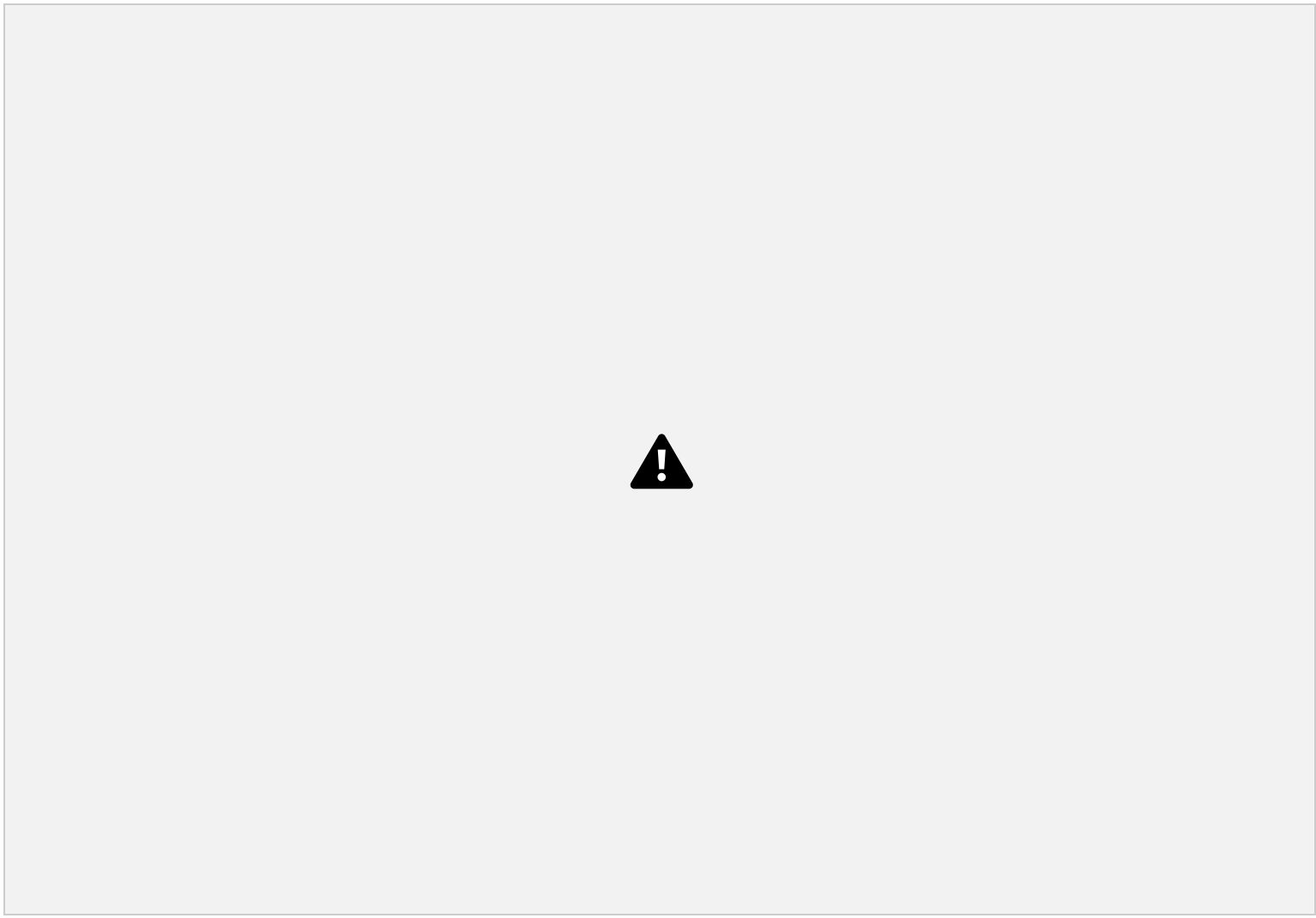
and UV light as illumination source

- Fluorochromes: substances that absorb short wavelength of light and emit longer wavelenght (visible)



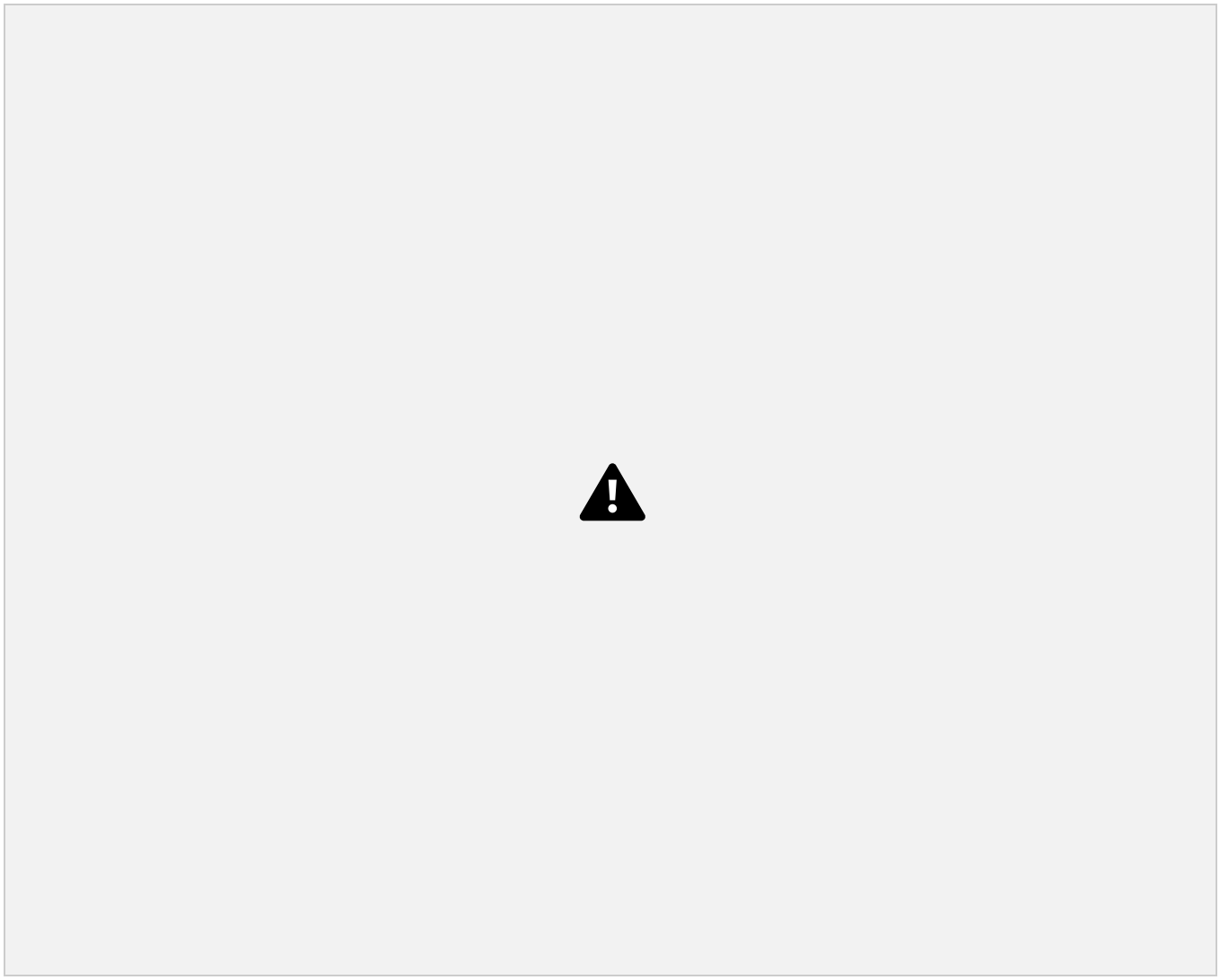


Fluorescent Microscopy



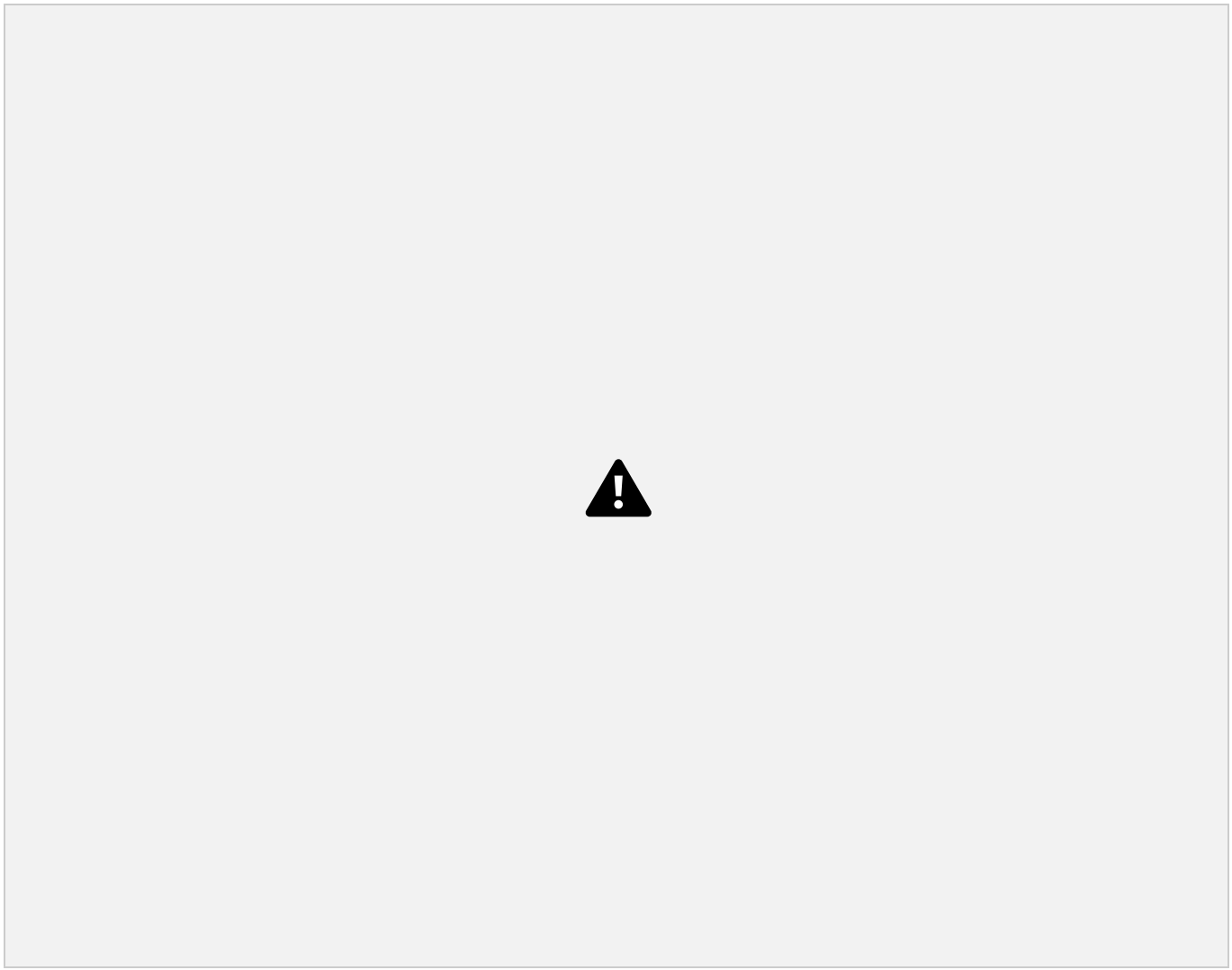


Fluorescent Microscopy



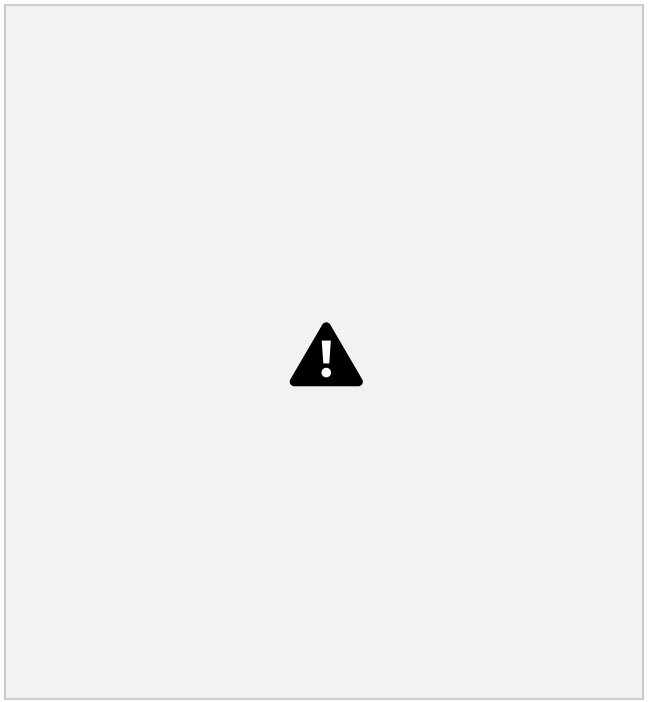


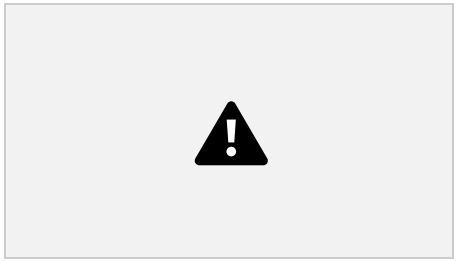
Fluorescent Microscopy

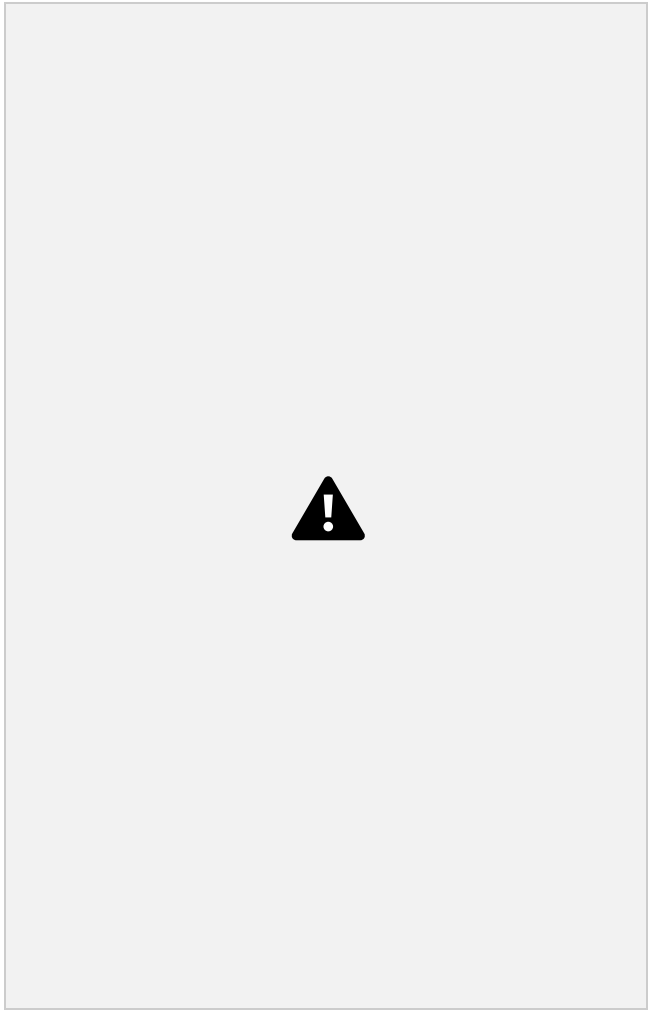




Fluorescent Microscopy



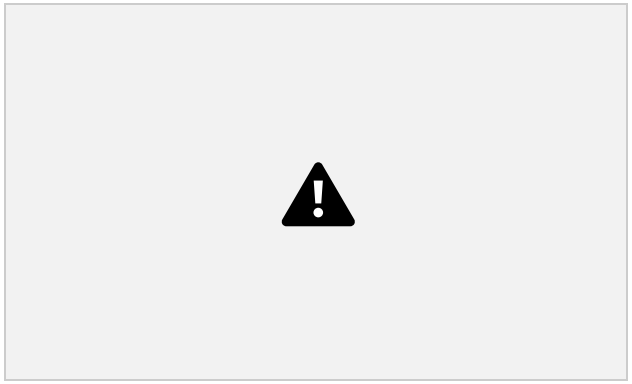






Transmission Electron Microscopy

Scanning Electron Microscopy



Staph aureus